

Indraneil Bardhan

Summary

Machine Learning Engineer with experience in deploying leading edge machine learning models on a wide range of cloud data platforms and unlocking potential hidden in large scale data systems. Eager to bring expertise and fresh ideas to innovative organization in challenging new role.

Skills

Focus Areas: MLOps, Predictive Modeling, Deep Learning, Recommender Systems, Data Engineering

Programming: Python, scikit-learn, h2o, MLLib, numpy, SciPy, geopandas, SHAP, matplotlib, plotly, R, tidyverse, ggplot2, mlr3, SQL

Frameworks: TensorFlow, PyTorch, PySpark, HIVE, BigQuery, elasticsearch, Google Cloud Platform (GCP), Amazon Web Services

Professional Experience

Senior Machine Learning Engineer - Climate/Bayer

Aug 2024 - Present

- Developed, validated and released an internal **Machine Learning Framework** for the **ML Research and Development group** to super-charge model training, testing, deployment and monitoring using **AWS SageMaker** and **Python**, reducing the time to production by 70%.
- Conducted **technical bootcamps** for domain knowledge experts to help them leverage Machine Learning best practices and adopt a **MLOps** mindset while building new AI Products for Climate FieldView with the goal of **improving operational effectiveness** and reducing the time to go to market.

Professional Experience

Machine Learning Engineer - Climate/Bayer

July 2021 - July 2024

- Productionized and deployed a **Bayesian Hybrid Pairing Recommendation** model using **PySTAN**, **Docker** and **AWS EC2** to deliver personalized seed hybrid recommendations into Climate FieldView, leading to an improvement in win rate by 45%.
- Productionized and deployed a **portfolio optimization** model for soy seed hybrids using **PySpark**, **Airflow** and **AWS EC2** to provide farmers with optimal seed hybrid portfolios.

Senior Data Scientist - r4 AI

May 2020 - July 2021

- Developed and evaluated **demand forecasting** models using **ARIMA**, **Holt-Winters** and **XGBoost** to help develop quarterly roadmaps to drive asset placement for a lumber manufacturer, leading to an overall lift of \$3M.
- Designed and deployed a **product recommendations** workflow to feed email promotions for an online sports retailer using **K-means** and **Collaborative Filtering**, doubling the conversion rate.
- Built and automated end to end **data pipelines** using **requests**, **BeautifulSoup**, **NLTK** and **cron** to identify high potential locations for a beverage manufacturer, increasing revenue from sales by 25%.

Data Scientist - r4 AI

April 2017 - May 2020

- Scaled a **location intelligence** model for a credit card provider, using **Rtree**, **shapely** and **geopandas** libraries, increasing coverage area by 80%.
- Prototyped a scalable **topic modeling** system on **PySpark** using **NLTK** and **LDA** on **gensim**, for an executive talent acquisition firm, improving time efficiency by 50%.
- Implemented **survival analysis** using **lifelines** and **Scikit-learn** for a manufacturing company, reducing unplanned stoppages by 35%.

Education

M. S. Information Science (Big Data Analytics) - University of Pittsburgh

2015 - 2016

B. Engg. Computer Engineering - University of Pune

2011 - 2015